

Title:

Inventory Management System Using C Programming

By - SUBHASHMITA SWAIN (590022427 _ b2)

Introduction:

An inventory management system is a software application designed to help organizations efficiently manage their stock of products, raw materials, and goods. In this project, we develop a simple yet functional Inventory Management System using the C programming language.

The program allows users to add, display, update, and delete product details such as product name, code, quantity, and cost. It serves as a foundation for understanding file handling, structures, and data management in C.

Problem Identification:

In many small businesses or stores, records of products are maintained manually, leading to errors, data loss, and difficulty in tracking stock levels.

Common problems include:

- Misplaced or lost inventory records
- Inaccurate product counts
- Difficulty updating stock information
- Lack of a centralized system for managing inventory
- To address these issues, a computerized system is needed that can store and retrieve product details efficiently.

Major Functionalities:**1. Add New Product:**

Allows the user to enter details such as product name, code, quantity, and cost.

2. Display All Products:

Shows all the stored product details in a well-formatted table view.

3. Search Product by Code:

Enables searching for a specific product using its unique code.

4. Update Product Information:

Allows modification of existing product details like cost or quantity.

5. Delete Product:

Removes a product record from the inventory.

6. Data Storage in File:

Product details are saved in a text file (inventory.txt) to ensure data persistence even after closing the program.

Usefulness of the Project:**The Inventory Management System offers several advantages:**

- Reduces manual errors and paperwork
- Simplifies stock tracking and updates
- Ensures data safety and easy access
- Helps small businesses manage their products efficiently

- Serves as a practical learning project for students to understand file handling, structures, and menu-driven programs in C

Conclusion:

This project demonstrates the importance of computerized record-keeping and data management in small-scale inventory systems. By implementing this C program, users can efficiently manage stock details, enhance accuracy, and improve productivity. The project can be further developed with advanced features such as graphical interfaces, barcode scanning, and database integration.